INTRODUCTION

Data have always been a key input for securities analysis and investment management, allowing investors to explore and exploit an abundance of information for their investment strategies. While this data-rich environment offers potentially tremendous opportunities for investors, turning data into useful information is not so straightforward.

This module provides a foundation for understanding important concepts that are an indispensable part of the analytical tool kit needed by investment practitioners, from junior analysts to senior portfolio managers. These basic concepts pave the way for more sophisticated tools that will be developed as the quantitative methods topic unfolds, which are integral to gaining competencies in the investment management techniques and asset classes that are presented later in the CFA curriculum.

This learning module focuses on how to summarize and analyze important aspects of financial returns, including key measures of central tendency, dispersion, and the shape of return distributions—specifically, skewness and kurtosis. The learning module finishes with a graphical introduction to covariance and correlation between two variables, a key concept in constructing investment portfolios to achieve diversification across assets within a portfolio.

Learning Module Overview

- Sample statistics—such as measures of central tendency, dispersion, skewness, and kurtosis—help with investment analysis, particularly in making probabilistic statements about returns.
- Measures of central tendency include the mean, the median and the mode, and specify where data are centered.
- The arithmetic mean is the sum of the observations divided by the number of observations. It is the most frequently used measure of central tendency.
- The median is the value of the middle item of observations, or the mean of the values of the two middle items, when the items in a set are sorted into ascending or descending order. Since the median is not influenced by extreme values, it is most useful in the case of skewed distributions.
- The mode is the most frequently observed value and is the only measure of central tendency that can be used with nominal or categorical data. A distribution may be unimodal (one mode), bimodal (two modes), trimodal (three modes), or have even more modes.
- Quantiles, as the median, quartiles, quintiles, deciles, and percentiles, are location parameters that divide a distribution into halves, quarters, fifths, tenths, and hundredths, respectively.
- A box and whiskers plot illustrates the distribution of a set of observations. The “box” depicts the interquartile range, the difference between the first and the third quartile. The “whiskers” outside of the “box” indicate the others measures of dispersion.
- Dispersion measures, such as the range, mean absolute deviation (MAD), variance, standard deviation, target downside deviation, and coefficient of variation, describe the variability of outcomes around the arithmetic mean.
- The range is the difference between the maximum value and the minimum value of the dataset. The range has only a limited usefulness because it uses information from only two observations.
- The MAD for a sample is the average of the absolute deviations of observations from the mean.
- The variance is the average of the squared deviations around the mean, and the standard deviation is the positive square root of variance. In computing sample variance, s^2, and sample standard deviation, s, the average squared deviation is computed using a divisor equal to the sample size minus 1.
- The target downside deviation, or target semideviation, is a measure of the risk of being below a given target.
- The coefficient of variation (CV) is the ratio of the standard deviation of a set of observations to their mean value. By expressing the magnitude of variation among observations relative to their average size, the CV allows for the direct comparisons of dispersion across different datasets. Reflecting the correction for scale, the CV is a scale-free measure, that is, it has no units of measurement.
- Skewness describes the degree to which a distribution is asymmetric about its mean. An asset return distribution with positive skewness has frequent small losses and a few extreme gains compared to a normal distribution. An asset return distribution with negative skewness has frequent small gains and a few extreme losses compared to a normal distribution. Zero skewness indicates a symmetric distribution of returns.

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions

(leptokurtic); a distribution with thinner tails than the normal distribution is referred to as thin-tailed (platykurtic). The kurtosis of a normal distribution is 3.

- The correlation coefficient measures the association between two variables. It is the ratio of covariance to the product of the two variables' standard deviations. A positive correlation coefficient indicates that the two variables tend to move together, whereas a negative coefficient indicates that the two variables tend to move in opposite directions. Correlation does not imply causation, simply association. Issues that arise in evaluating correlation include the presence of outliers and spurious correlation.

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